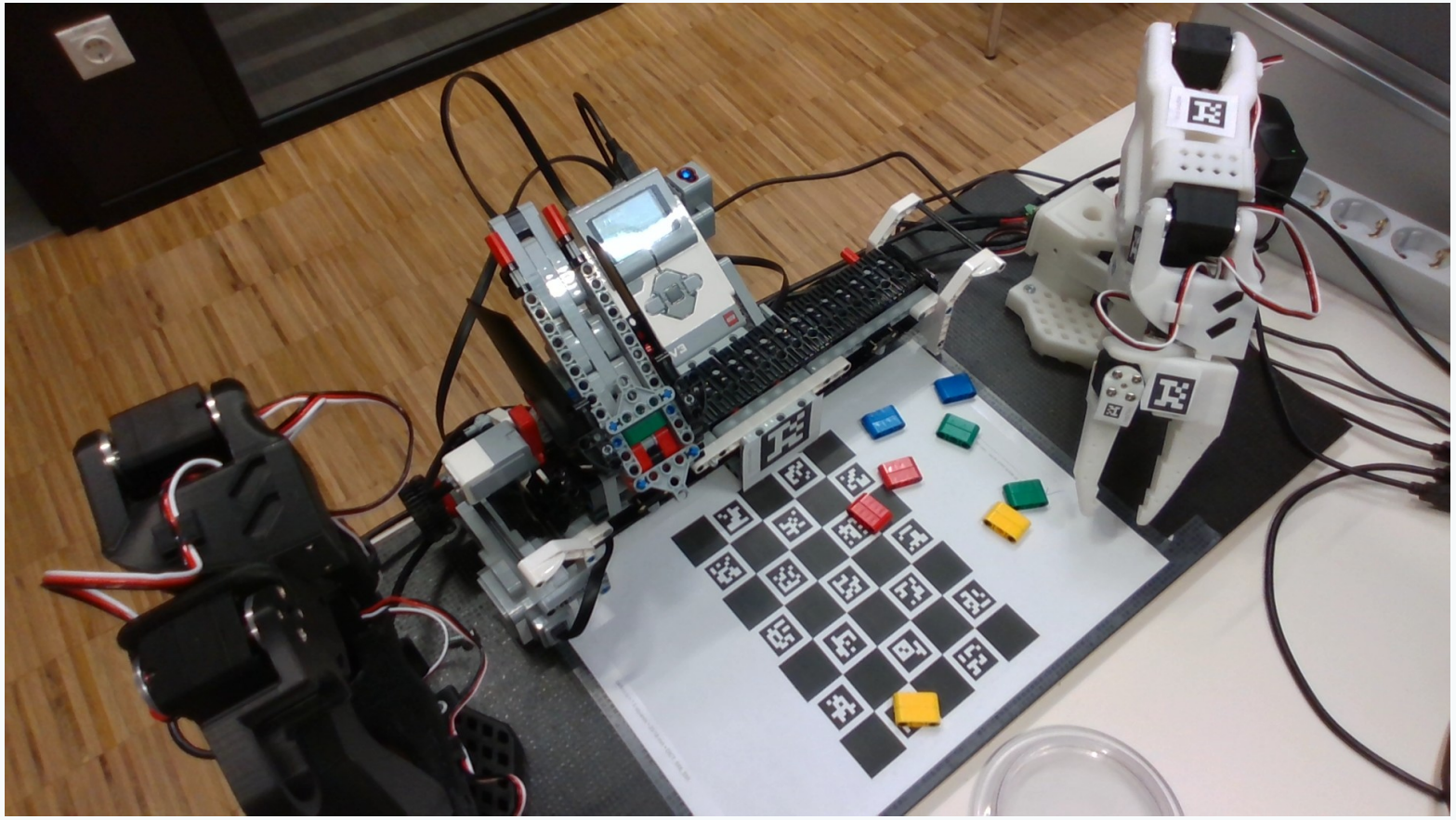
# Our fixed-fixture factory



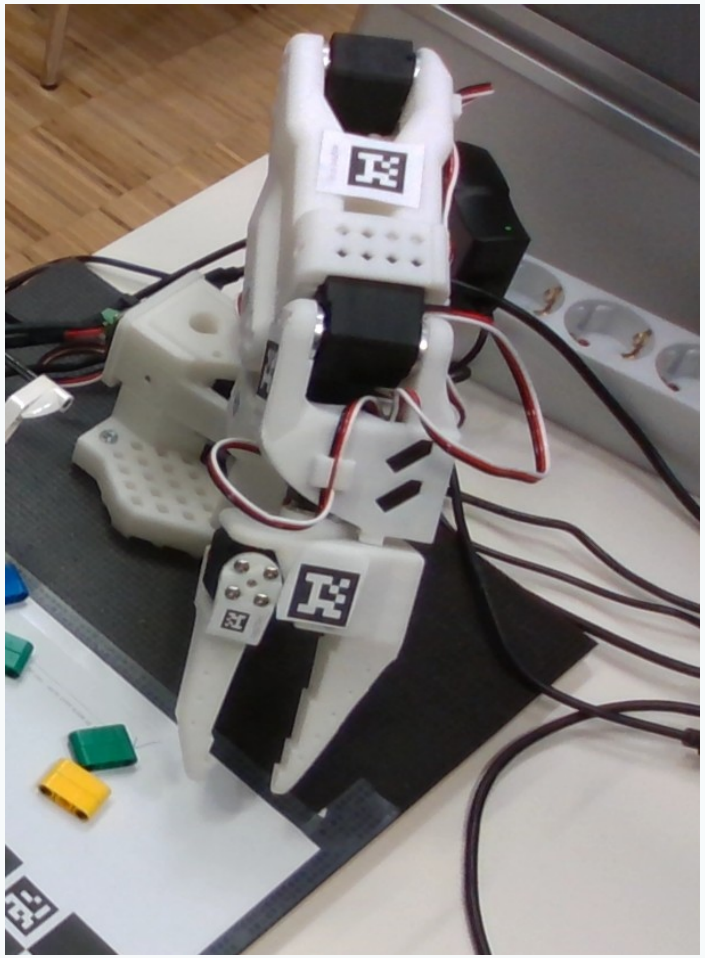
**Dispense one object**

Carry it to a stopped pickup fixture

Pick, place, release and return

Actual station photograph · one fixed pickup and one fixed destination

# Measure decide act check



**Sensor measures**

↓

Program decides

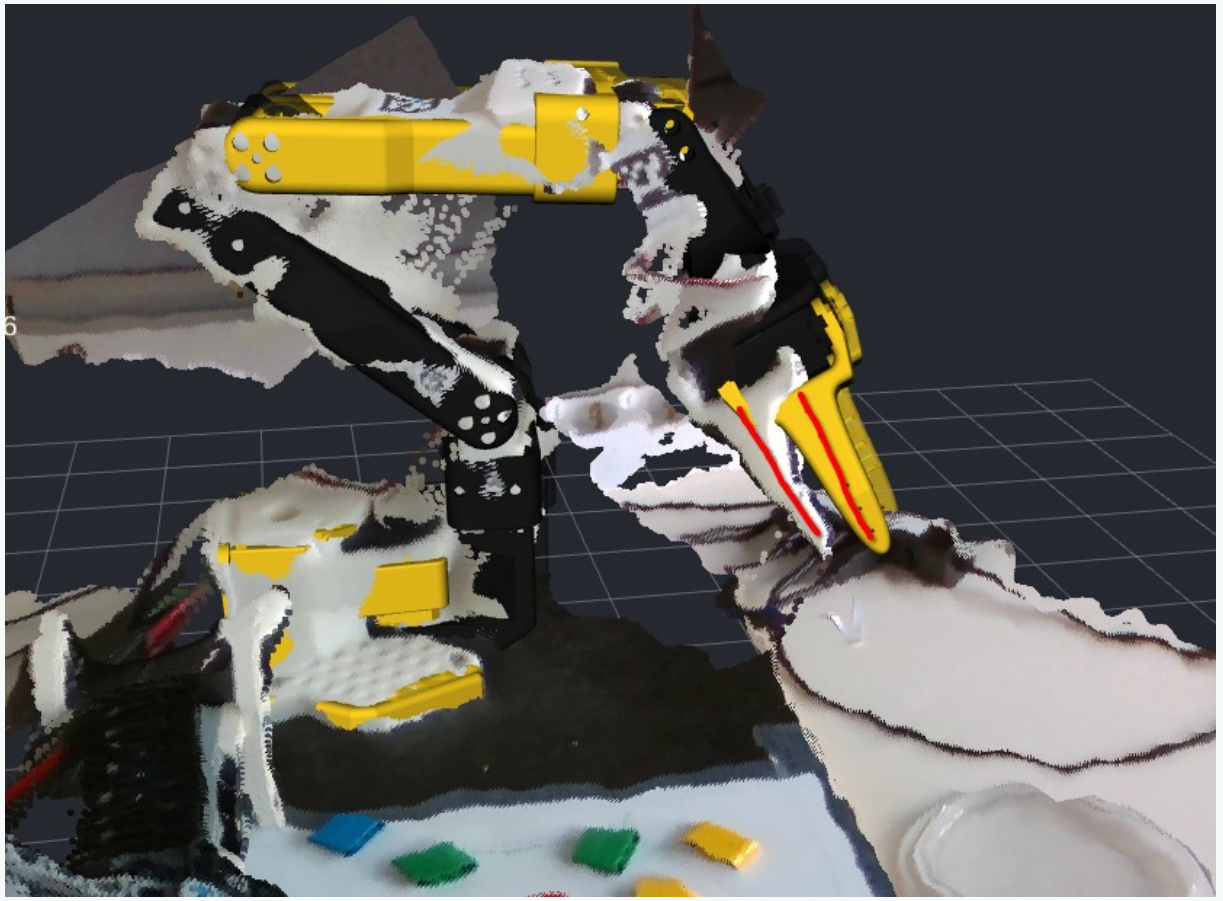
↓

Actuator moves

↓

Inspection checks the result

# Robot and screen twin



**Leader: handle**

Follower: physical arm

Twin: screen model

# Put the steps in order

1 Dispense

↓

**2 Conveyor to fixture and stop**

↓

3 Approach and grasp

↓

4 Lift and place

↓

5 Release and return

# An observation can be wrong

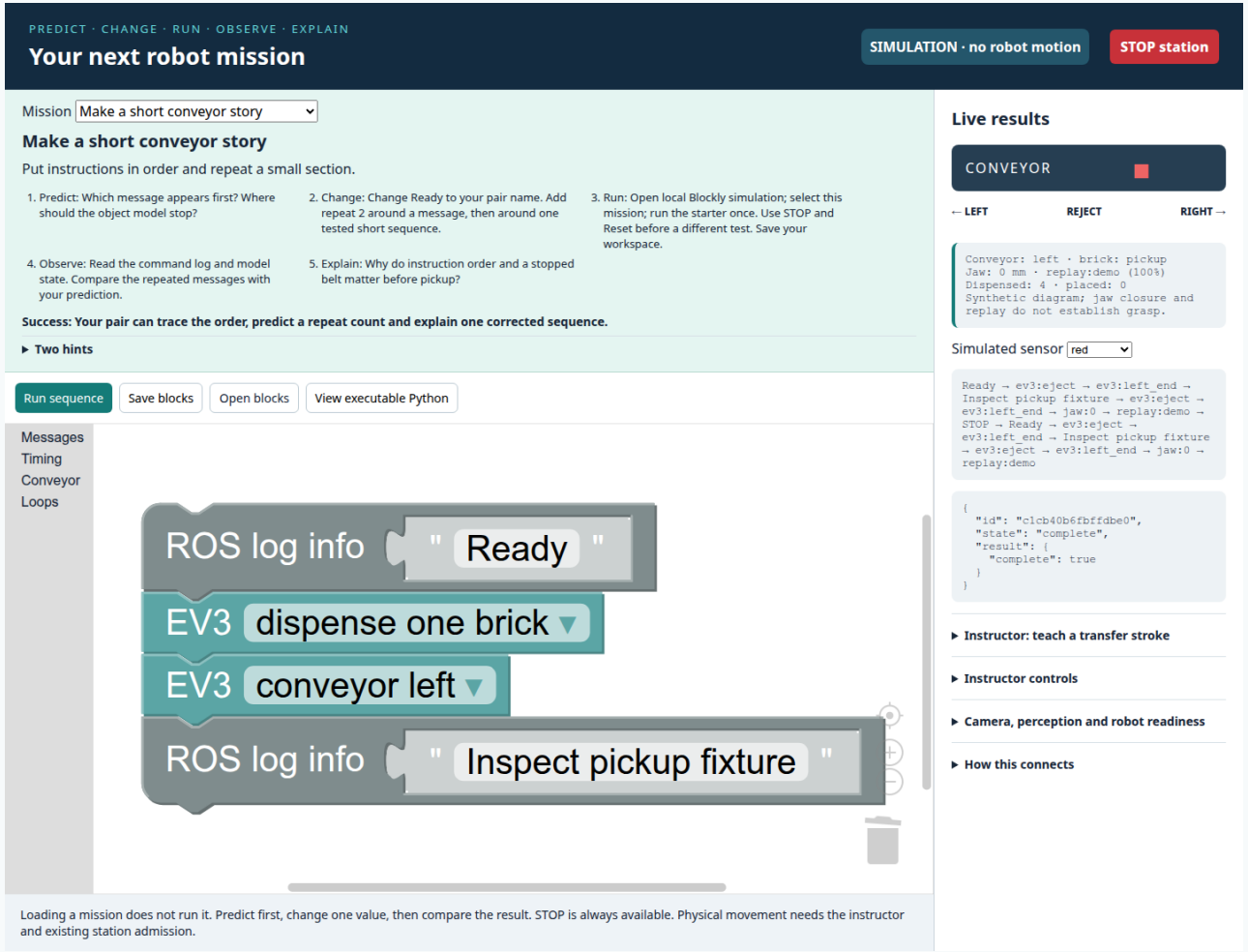


**Actual object: red**

Reported label: blue (altered example)

Missing report: pause and inspect

# Blocks run in order




Read from the top

Repeat the blocks inside

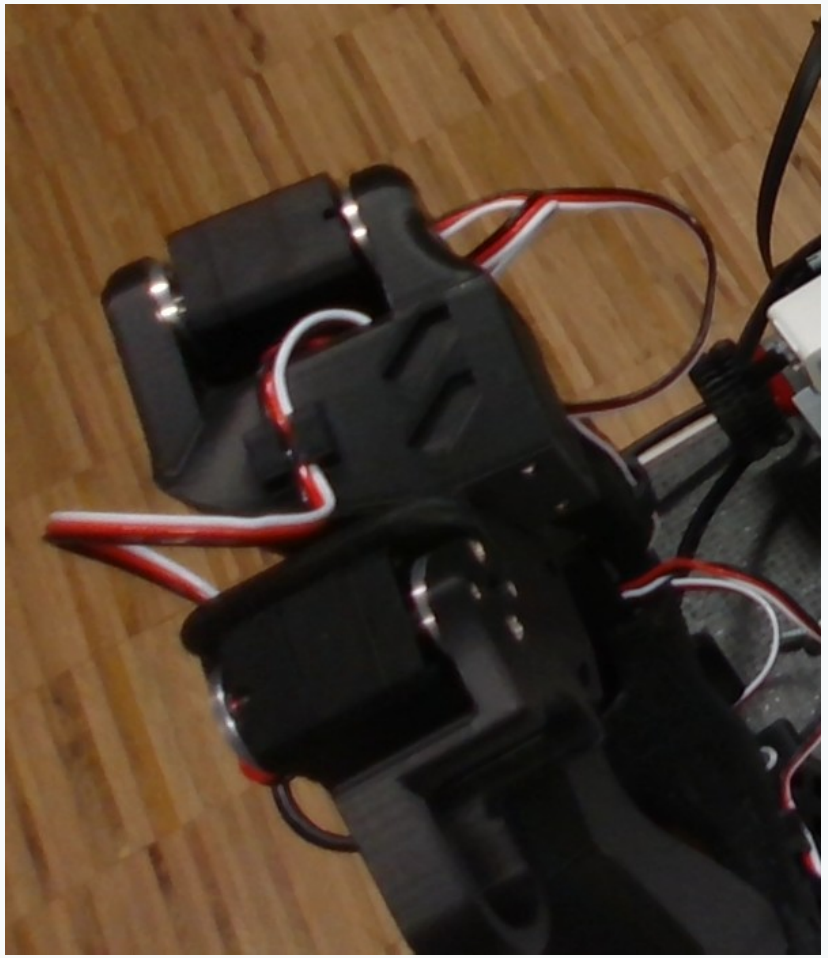
**Predict before Run**

# Your first Python change

```
TEAM = "Pair A"
REPEATS = 3
print("Hello", TEAM)
print("Count", REPEATS)
```

**Predict the output, then change one value.**

# Teach inspect return replay



**Stroke: one saved movement**

Pose: arm arrangement at start/end

Jaw state: gripper opening

Calibration: measured references

# Make a short conveyor story

## Mission 1 · 25 minutes total · Predict and run

| | |
|---|---|
| **Worked example** | Ready → dispense → belt to pickup → inspect. Simulation is a model. |
| **Predict** | Which instruction runs first? Where should the model stop? |
| **Change** | Change Ready to your pair name. Repeat one tested message twice. |
| **Run** | Run the local Blockly starter once. Use Reset simulation between experiments. |

# Make a short conveyor story

## Mission 1 · 25 minutes total · Check and explain

| | |
|---|---|
| **Observe** | Compare command order, model position and repeated messages. |
| **Explain** | Why must the conveyor stop before the arm approaches? |
| **Success** | Trace the order, predict repeat count and explain one correction. |
| **Hints** | Read connected blocks from top to bottom.<br>Repeat runs every block inside it. |

Optional: Try a wrong order in simulation and explain what changed.

# Separate the object from its report

## Mission 2 · 20 minutes total · Predict and run

| | |
|---|---|
| **Worked example** | Actual/report (invented reports): 1 red/red; 2 blue/None; 3 red/blue. |
| **Predict** | Which report is missing? Which disagrees with the actual object? |
| **Change** | Change one report to None; then change a known report to a wrong color. |
| **Run** | Human-labelled photo: handbook, observation mission; ZIP notebooks/assets/photo-examples.jpg. Reports invented. |

# Separate the object from its report

## Mission 2 · 20 minutes total · Check and explain

| | |
|---|---|
| **Observe** | Record actual object and reported label in separate columns. |
| **Explain** | Why can a fresh, familiar label still be wrong? |
| **Success** | Identify one missing and one wrong report; choose pause and inspect. |
| **Hints** | Compare actual and reported one row at a time.<br>A known word is still only a report. |

Optional: Test green without adding a rule.

# Change a name and a number

**Mission 3 · 30 minutes total · Predict and run**

| | |
|---|---|
| **Worked example** | TEAM stores text; REPEATS stores a number. Indentation defines the loop. |
| **Predict** | What greeting and how many object lines will appear? |
| **Change** | Change TEAM. Run. Then predict and change REPEATS to 2. |
| **Run** | Run notebook from the top. In Blockly choose "Change a name and a number", then View executable Python. Simulation only. |

# Change a name and a number

**Mission 3 · 30 minutes total · Check and explain**

| | |
|---|---|
| **Observe** | In "Change a name and a number", use View executable Python. Compare its rows with program.program. |
| **Explain** | Which edit changed text? Which changed the number of repetitions? |
| **Success** | Both partners make and explain one edit; share the same validated commands. |
| **Hints** | Quotes surround text, not the repeat number.<br>Keep four spaces before the repeated print. |

Optional: Change a message in Python and Blockly; compare both traces.

# Watch teach inspect return replay

## Mission 4 · 15 minutes total · Predict and run

| | |
|---|---|
| **Worked example** | Stroke: one saved movement. Pose: start/end arrangement. Jaw state: gripper opening. Calibration: measured references. |
| **Predict** | Will a replay start safely from any pose? What if the jaw state changed? |
| **Change** | Change one invented angle after predicting. Instructor owns physical modes. |
| **Run** | Instructor: teach, stop recording, inspect recording details/path, return to validated start, then replay separately. Locate STOP. |

# Watch teach inspect return replay

## Mission 4 · 15 minutes total · Check and explain

| | |
|---|---|
| **Observe** | Identify start, endpoint, jaw state and validation evidence. |
| **Explain** | Why separate teaching from replay? Why reteach after calibration changes? |
| **Success** | Explain teach → inspect → return → replay, and locate STOP. |
| **Hints** | A saved stroke does not plan from every start pose.<br>A matching screen does not prove a physical grasp. |

Optional: Compare relative 35° with absolute 15°. Invented arithmetic sends no motion.

# Explore three stations

## Mission 5 · 45 minutes total · Predict and run

| | |
|---|---|
| **Worked example** | SIMULATION: changed leader now 10° → 15°; predicted and calculated relative 30 + 15 - 10 = 35°. No physical joint was measured. |
| **Predict** | Predict one direction, report or instruction order at this station. |
| **Change** | Change one angle, report or message. Swap driver and observer. |
| **Run** | 15 minutes: 2 predict, 7 change/run, 4 observe/explain, 2 transition. |

# Explore three stations

**Mission 5 · 45 minutes total · Check and explain**

| | |
|---|---|
| **Observe** | Record one observation and label it PAPER, SIMULATION or PHYSICAL. |
| **Explain** | Which part measures, decides or acts? What can your evidence establish? |
| **Success** | Every pair visits all three positions and records prediction plus evidence. |
| **Hints** | The twin is a model updated from a stream.<br>Use the same short sequence at the offline desk. |

Optional: Compare fresh with stale without guessing a missing angle.

# Dispense carry pick and place

**Mission 6 · 45 minutes total · Predict and run**

| | |
|---|---|
| **Worked example** | Dispense → conveyor/stop → fixed_pick_place. Missing qualification blocks hardware. |
| **Predict** | Where must the object and arm be before each step? What should stop the run? |
| **Change** | Reorder paper cards. Use the named skill fixed_pick_place in simulation. |
| **Run** | Rotate: supervised integration, offline rehearsal, failure/evidence desk. |

# Dispense carry pick and place

**Mission 6 · 45 minutes total · Check and explain**

| | |
|---|---|
| **Observe** | Inspect dispense, arrival, grasp, lift, release and return separately. |
| **Explain** | Which measurement supports each step? What remains unverified? |
| **Success** | Explain the full sequence and label synthetic rehearsal versus observed transfer. |
| **Hints** | Stop belt; match the validated start and jaw state.<br>Missing qualification blocks hardware; ask instructor. |

Optional: After fixed-fixture success, discuss separately validated color destinations.

# Find one failure and test a repair

## Mission 7 · 20 minutes total · Predict and run

| | |
|---|---|
| **Worked example** | present, missing, present: pause after one inspected simulated object. |
| **Predict** | What should happen at the missing middle report? |
| **Change** | Choose one failure and one repair. Reset before the repeat check. |
| **Run** | Reproduce on paper or in simulation. Repair and rerun the same input. |

# Find one failure and test a repair

**Mission 7 · 20 minutes total · Check and explain**

| | |
|---|---|
| **Observe** | Record failure message, stopped state and the repaired trace. |
| **Explain** | Did the test detect the problem? What could still fail on hardware? |
| **Success** | Show a failed prediction, one repair and a repeated check with limits. |
| **Hints** | Change one cause so you can explain its effect.<br>One passing run can still hide a wrong known report. |

Optional: Try three reds and one blue to expose swapped extension rules.

# Show one prediction and result

## Mission 8 · 10 minutes total · Predict and run

| | |
|---|---|
| **Worked example** | SIMULATION: repeat count 1→2. Predicted Ready twice; log showed Ready twice. This checks messages, not physical arrival or grasp. |
| **Predict** | Which saved output best supports your claim? |
| **Change** | Choose one change and one observation for the demonstration. |
| **Run** | 1 minute prepare; 3 × 2-minute pair demos; 3 minutes reflection. |

# Show one prediction and result

**Mission 8 · 10 minutes total · Check and explain**

| | |
|---|---|
| **Observe** | Show the trace or worksheet; label paper, simulation or physical evidence. |
| **Explain** | What next test would increase confidence? |
| **Success** | All three pairs demonstrate; both partners speak and name a limitation. |
| **Hints** | Read your prediction before showing the result.<br>A trace shows order; arrival needs physical evidence. |

Optional: Explain wrong known color versus missing observation.